

RAG

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1 vector databases

1.1 Explain in your own words what a vector database is and how its different from a traditional database like MySQL or PostgreSQL.

in vector database the data we need is collected by searching similar words in the database. when it comes to MySQL and PostgreSQL if we need the data it will be given by using the index to find the data. vector database can search for vectors based on similarities.

1.2 Take any sentence of your choice. Describe what happens to it step by step from raw text to being stored in a vector database. Name each stage.

consider the statement "i love shawarma"

initially the text is just raw information, there is no use of this information. this is the stage where the text is in its raw state.

in the next stage we will use chunking to convert the text into smaller parts so that chunk becomes a separate piece of information which can be used.

in the next stage we will use embedding model which will convert the text into numbers. this is useful because it helps RAG store the words with meanings.

in the next stage the data will be stored in the vector database. in the storage process the embeddings, and any other information related to it will be stored in the vector database. with the help of indexing the database will be organised in a way that the data is obtained efficiently .

for example the statement "i love shawarmas" if we input anything related to shawarmas any metadata or information related to shawarmas can be obtained easily.

1.3 Why can't we just use a normal database index (like a B-tree) to search through embeddings? Explain the core difference between exact lookup and similarity search.

mostly the reason is because when we talk about B-trees, they tend to give accurate answers, whereas when we are looking for an embedding we need similarity. there might be situations where we don't need the exact embedding but something related to it and that's when we would use normal embeddings. similarity search gives us exact value of an embedding which might cause the LLM to give wrong answers.

1.4 Explain why a vector database is essential for RAG specifically. What would break if you removed it from the pipeline?

vector database is necessary because it helps the LLM find similar texts with the help of embeddings, the vector database allows the RAG to find similar words for the embedding and it helps retrieve similar information. we can't do that if there is no vector database. it would take the llm a long time to scan the entire file again and again in order to get the information out.

2 Advanced RAG

2.1 You're building a RAG system for a company's product manuals. These manuals have text instructions along with diagrams and photos showing how to assemble the product. A user asks, "How do I attach the side panel," and the answer is shown entirely in a diagram, with no accompanying text. Explain how you would approach this with multimodal RAG. What would go wrong if you used a text-only RAG pipeline here?

if the issue occurs, i would first use the OCR model to convert the image to speech, which will help the worker understand the process even better when he gets the proper text. if i used the text-only RAG pipeline, the major issue would be about where the parts need to go, even if we have the instructions, we would still need some description about how a part looks while assembling the product. text won't be effective while describing the part also we need a final look about how the product should look like.

2.2 Briefly compare the two approaches to multimodal retrieval covered in lecture: unified embedding space vs separate indexes with fusion. Which one would you pick for the product manual scenario above, and why?

Multimodal RAG can process both text and visual content, such as diagrams and photographs, from product manuals. OCR can extract written labels from diagrams, while a vision-language model can describe the visual information. This text can be converted into embeddings and stored in a vector database along with the original images.

When a user asks a question, the system retrieves relevant text or images and provides them to the LLM to generate an answer. A text-only RAG system may fail when important information is available only in diagrams or photographs.